

PUNEET MATHUR

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EDUCATION

University of Maryland, College Park

Ph.D. and M.S. in Computer Science; GPA: 3.97/4.0

Advised by [Dr. Dinesh Manocha](#)

Netaji Subhas Institute of Technology (NSIT)

B.E. in Computer Engineering; GPA: 3.8/4.0

College Park, MD, USA

Aug 2019 – Dec 2023

New Delhi, India

Aug 2014 – May 2018

PROFESSIONAL EXPERIENCE

Adobe Research

Research Scientist, Agentic Intelligence Research (AIR Lab)

San Jose, CA, USA

Sept 2023 – Present

- **Full-duplex audio language modeling:** Building robust evaluation benchmarks for full-duplex speech agents (document grounding, hallucination, and instruction-following), and post-training full-duplex audio LMs to think, reason, and retrieve during conversation for grounded question answering; mitigating latency via learned audio fillers and GRPO post-training for reasoning and retrieval, alongside efficient inference optimizations for omni-modal models.
- **Partial-policy gradients for RL in LLMs:** Novel credit assignment technique for offline policy gradients that scales sub-horizon planning for multi-step LLM trajectories, with scaling laws for policy gradient finetuning and improved long-context persona-following in LLM roleplaying.
- **Reasoning-based LLM clustering via RL/GRPO:** Post-trained Qwen models via GRPO to reason over complex user queries for organizing large text collections, improving query-aware clustering by +14% V-score (at par with GPT-o3).
- **Roleplaying agents:** Multi-agent orchestration layer with planning, behavior alignment, and role verification for consistent specialist personas; RL reward modeling and curriculum learning for tutor agents.
- **Self-evolving agents:** Led development of a self-evolving tutor agent using TextAdam, a conversational prompt optimization technique yielding a 23% gain in instruction following over TextGrad and GEPA.
- **Agentic memory and continual learning:** Novel memory architectures that let LLM agents learn from past interactions and reduce recurring errors; deployed in the Common Agentic Framework and used by 5+ Adobe products.
- **LLM attribution and RAG:** Post-hoc attribution agents providing citations for QA systems over document text, tables, charts, and visual artifacts via multimodal retrieval and agentic verification; improved table attribution in Acrobat AI Assistant by 12%.

RESEARCH INTERNSHIPS

Microsoft Research

Research Scientist Intern, Visual Document Understanding

Redmond, WA, USA

May 2023 – Aug 2023

- Developed a Transformer-based model for chart information extraction that recovers chart structure and infills underlying data values, improving F1 score by 15% over prior methods.

Meta AI (Facebook AI)

Research Scientist Intern, NLP & Speech Team

Menlo Park, CA, USA

Sept 2022 – May 2023

- AI for speech recognition and language model personalization: retrieval-augmented language modeling for NLP+speech, generative NLP for factual consistency and domain adaptation, and embedding-based retrieval for query understanding, improving word-error rate in ASR across several datasets. Papers accepted to EMNLP 2023 and Interspeech 2023.

Adobe Research

Research Scientist Intern, Document Intelligence Lab

San Jose, CA, USA

June 2022 – Aug 2022

- Automated document editing based on natural language prompt instructions. Paper accepted to AAAI 2023.

Verisk AI

AI/ML Research Intern

New Jersey, USA

Jan 2022 – June 2022

- Natural language inference on legal documents using optimal evidence selection. Paper accepted to EMNLP 2022.

Adobe Research

San Jose, CA, USA (Remote)

Research Scientist Intern, Document Intelligence Lab

June 2021 – Aug 2021

- End-to-end extraction of hierarchical document structures from semi-structured forms and invoices. Paper accepted at WACV 2023.

Dataminr Inc.

New York, USA

AI Research Intern

June 2020 – Aug 2020

- Duplicate detection of news headlines as a tiered (event-entity-phrase level) paraphrase identification task, deployed as an internal system.

SELECTED PUBLICATIONS ON FULL-DUPLEX AUDIO & SPEECH - [ALL PUBLICATIONS](#)

1. Duplex-R1: Full-Duplex Audio LMs that Reason, Retrieve, and Speak While Searching
Puneet Mathur, Nedim Lipka, Zeyu Jin, Dinesh Manocha
In submission, 2026 (*Coming soon*)
2. DuplexSpeechBench–Document Grounding: Benchmarking Document Grounding and Hallucinations in Full-Duplex Voice Agents
Puneet Mathur, Nedim Lipka, Zeyu Jin, Dinesh Manocha
In submission, 2026 (*Coming soon*)
3. DuplexSpeechBench–IFEval: Instruction Following Evaluation for Full-Duplex Voice Agents
Puneet Mathur, Manan Suri, Dinesh Manocha
In submission, 2026 (*Coming soon*)
4. Omni2LoRA: Coherence-Preserving Parametric Memory for Efficient Omni Language Models
Puneet Mathur, Manan Suri, Dinesh Manocha
In submission, 2026 (*Coming soon*)

SELECTED PUBLICATIONS ON AGENTIC AI & REINFORCEMENT LEARNING - [ALL PUBLICATIONS](#)

1. [ARGUS: Structured Uncertainty guided Clarification for LLM Agents](#)
Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, Dinesh Manocha
ACL 2026
2. [Partial Policy Gradients for RL in LLMs](#)
Puneet Mathur, Branislav Kveton, Subhojyoti Mukherjee, Viet Dac Lai
In submission, 2026
3. [Cluster-R1: Large Reasoning Models Are Instruction-following Clustering Agents](#)
Peijun Qing, Puneet Mathur, Nedim Lipka, Varun Manjunatha, Ryan A. Rossi, Franck Dernoncourt, Saeed Hassanpour, Soroush Vosoughi
In submission, 2026

SELECTED PUBLICATIONS ON DOCUMENT INTELLIGENCE & RAG - [ALL PUBLICATIONS](#)

1. [ChartEval: LLM-Driven Chart Generation Evaluation Using Scene Graph Parsing](#)
Kanika Goswami, Puneet Mathur, Franck Dernoncourt, Ryan A. Rossi, Vivek Gupta, Dinesh Manocha
AACL 2025
2. [ChartLens: Fine-grained Visual Attribution in Charts](#)
Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, Dinesh Manocha
ACL 2025
3. [Follow the Flow: Fine-grained Flowchart Attribution with Neurosymbolic Agents](#)
Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, Vivek Gupta, Dinesh Manocha
EMNLP 2025
4. [VisDoM: Multi-Document QA with Visually Rich Elements Using Multimodal Retrieval-Augmented Generation](#)
Manan Suri, Puneet Mathur, Franck Dernoncourt, Kanika Goswami, Ryan A. Rossi, Dinesh Manocha
NAACL 2025

5. [MoDS: Moderating a Mixture of Document Speakers to Summarize Debatable Queries in Document Collections](#)
Nishant Balepur, Alexa Siu, Nedim Lipka, Franck Deroncourt, Tong Sun, Jordan Lee Boyd-Graber, *Puneet Mathur*
NAACL 2025
6. [MATSA: Multi-Agent Table Structure Attribution](#)
Puneet Mathur, Alexa Siu, Nedim Lipka, Tong Sun
EMNLP 2024
7. [DocEdit-v2: Document Structure Editing Via Multimodal LLM Grounding](#)
Manan Suri, *Puneet Mathur*, Franck Deroncourt, Rajiv Jain, Vlad I Morariu, Ramit Sawhney, Preslav Nakov, Dinesh Manocha
EMNLP 2024
8. [DocPilot: Copilot for Automating PDF Edit Workflows in Documents](#)
Puneet Mathur, Alexa Siu, Varun Manjunatha, Tong Sun
ACL 2024
9. [DocEdit: Language-guided Document Editing](#)
Puneet Mathur, Rajiv Jain, Jiuxiang Gu, Franck Deroncourt, Dinesh Manocha, Vlad Morariu
AAAI 2023
10. [TIMERS: Document-level Temporal Relation Extraction](#)
Puneet Mathur, Rajiv Jain, Franck Deroncourt, Vlad Morariu, Quan Hung Tran, Dinesh Manocha
ACL 2021
11. Several other works on document temporal-relation extraction, layout understanding, and text-to-SQL robustness, published at EMNLP/NAACL/WACV/ACL: [see full list here](#)
Puneet Mathur et al.

SELECTED PUBLICATIONS ON MULTIMODAL AI - [ALL PUBLICATIONS](#)

1. [PersonaLM: Language Model Personalization via Domain-distributed Span Aggregated K-Nearest N-gram Retrieval Augmentation](#)
Puneet Mathur, Zhe Liu, Ke Li, Yingyi Ma, Gil Keren, Zeeshan Ahmed, Dinesh Manocha, Xuedong Zhang
EMNLP 2023 (Findings)
2. [DOC-RAG: ASR Language Model Personalization with Domain-Distributed Co-occurrence Retrieval Augmentation](#)
Puneet Mathur, Zhe Liu, Ke Li, Yingyi Ma, Gil Keren, Zeeshan Ahmed, Dinesh Manocha, Xuedong Zhang
COLING 2024
3. [DocLayoutTTS: Dataset and Baselines for Layout-informed Document-level Neural Speech Synthesis](#)
Puneet Mathur, Franck Deroncourt, Quan Hung Tran, Jiuxiang Gu, Ani Nenkova, Vlad Morariu, Rajiv Jain, Dinesh Manocha
Interspeech 2022
4. [MONOPOLY: Financial Prediction from MONetary POLicY Conference Videos Using Multimodal Cues](#)
Puneet Mathur, Atula Tejaswi Neerkaje, Malika Chhibber, Ramit Sawhney, Fu-Ming Guo, Franck Deroncourt, Sanghamitra Dutta, Dinesh Manocha
ACM Multimedia 2022
5. [Affect2MM: Affective Analysis of Multimedia Content Using Emotion Causality](#)
Trisha Mittal, *Puneet Mathur*, Aniket Bera, Dinesh Manocha
CVPR 2021
6. Several other works on speech emotion recognition, financial risk forecasting, and early vision/social-media NLP, published at EMNLP/Interspeech/ICASSP/AAAI: [see full list here](#)
Puneet Mathur et al.

SKILLS

Programming Languages: Python, C++ , Java, SQL

Data Science Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Jupyter Notebook, NumPy, Pandas

Software Technologies: Flask, Kafka, AWS, Git